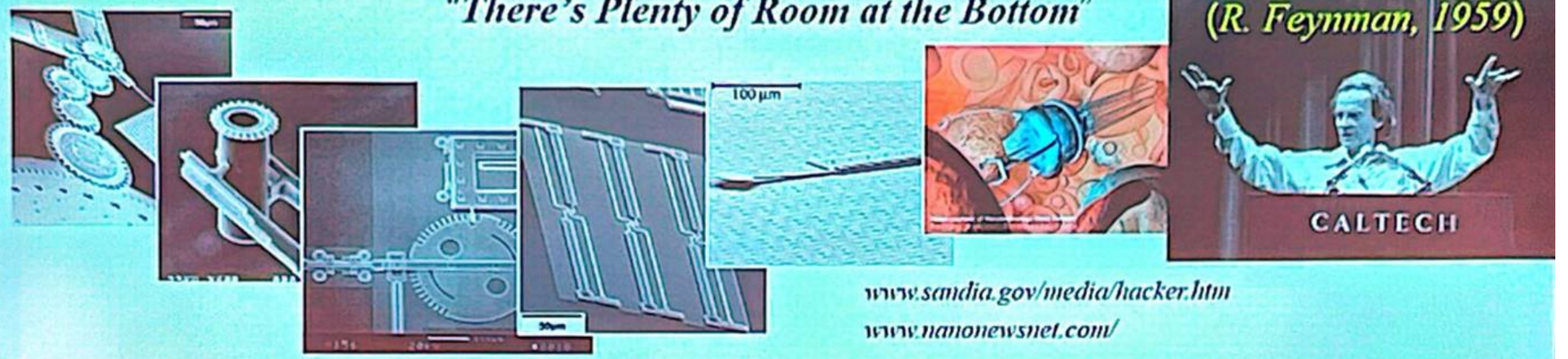
Nanomechanics

"There's Plenty of Room at the Bottom"



www.sandia.gov/media/hacker.htm

www.nanonewsnet.com/

~ a branch of nanoscience studying fundamental mechanical (elastic, thermal and kinetic) properties of physical systems at the nanometer scale. … emerged on the crossroads of **biophysics, classical mechanics, solid-state physics, statistical mechanics, materials science, and quantum chemistry**.

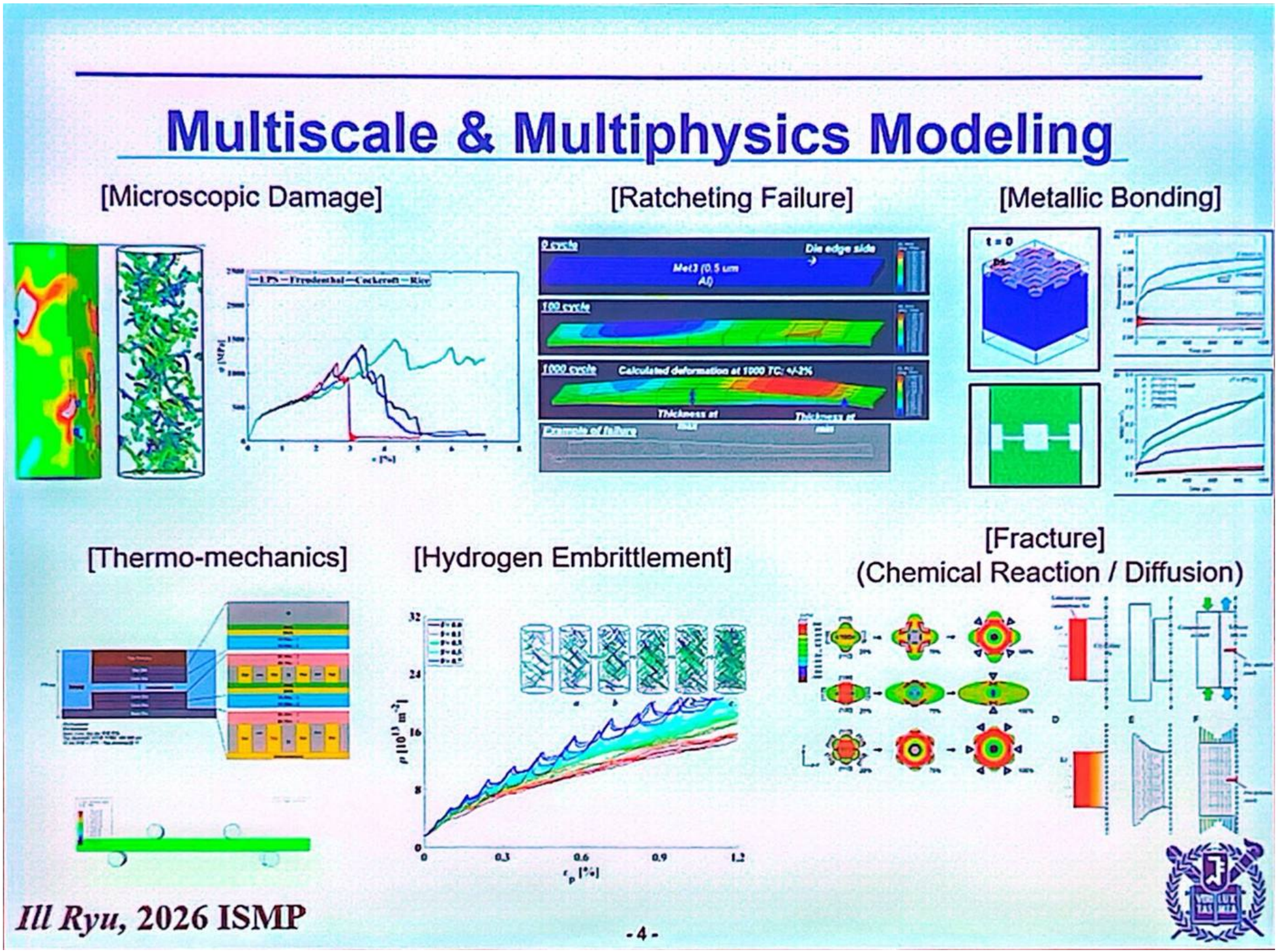
Why does it matter? ***- Size dependent mechanical properties***
- Intermittency, Stochasticity

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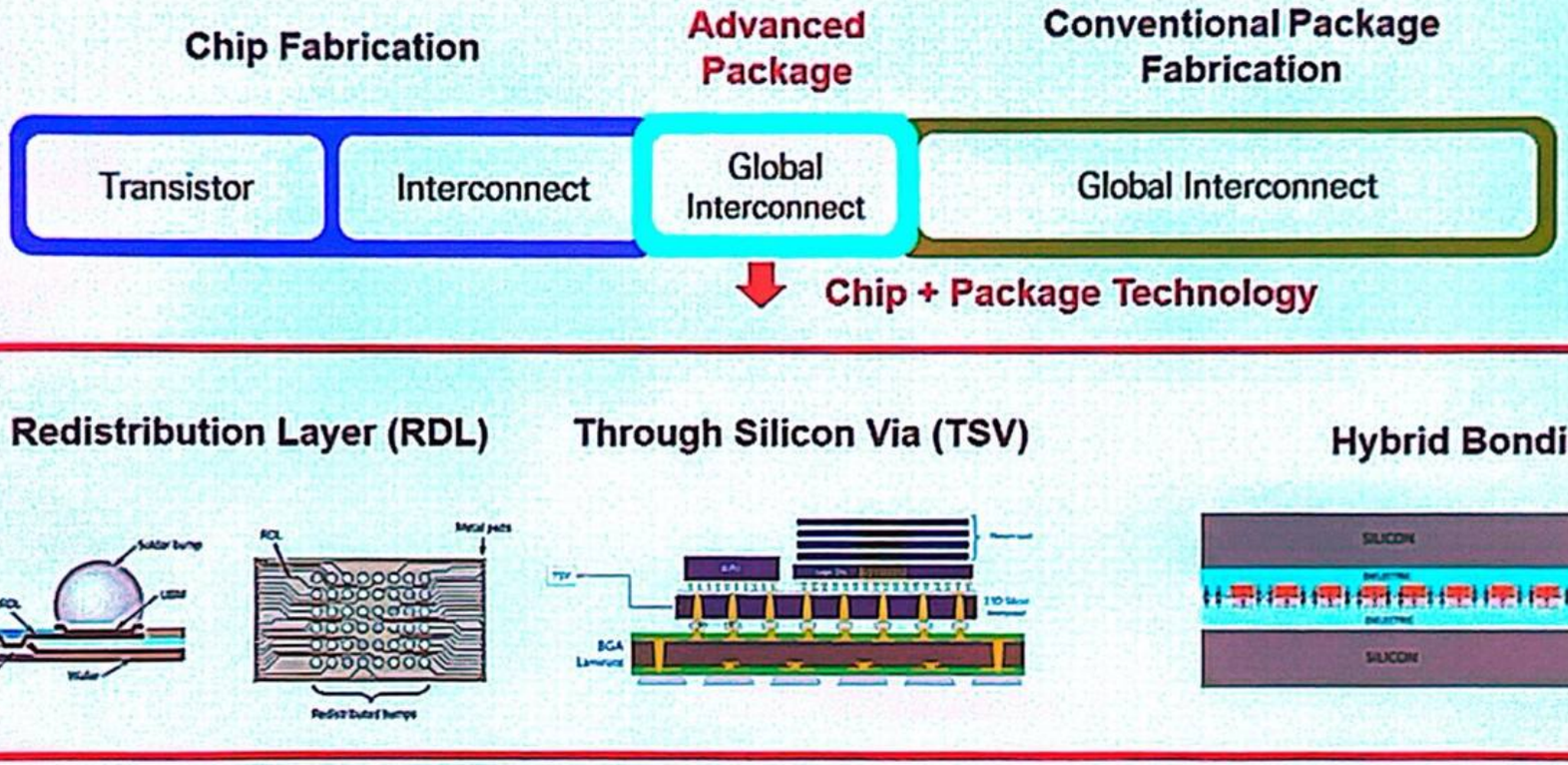
-2-

T32 다중 스케일 모델로 보는 패키지 파괴

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Challenges in Advanced Package Reliability



- Advanced packaging technology converges chip and conventional packaging technologies. Rapid shrinkage of advanced packaging pitch size may lead to various reliability issues.

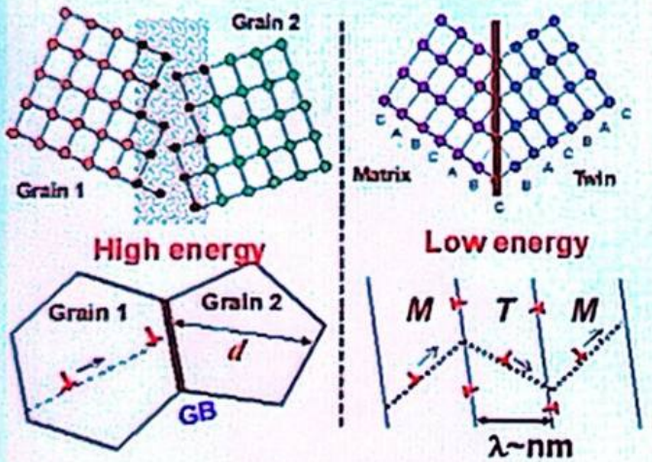
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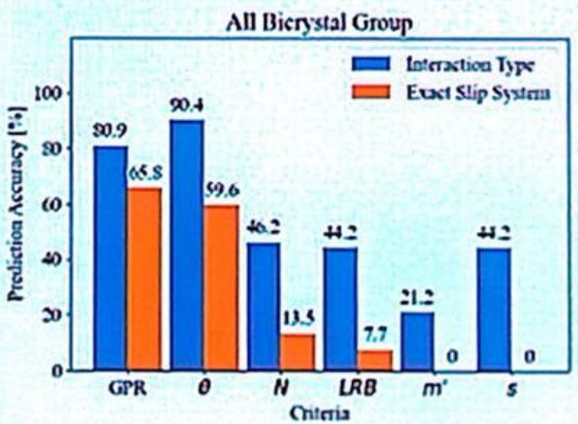
Material Informatics

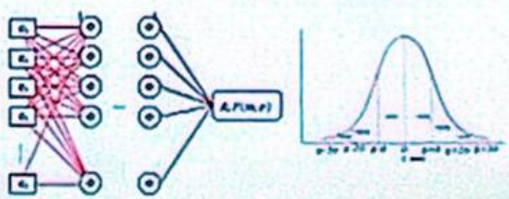
[Incoherent vs Coherent GBs]

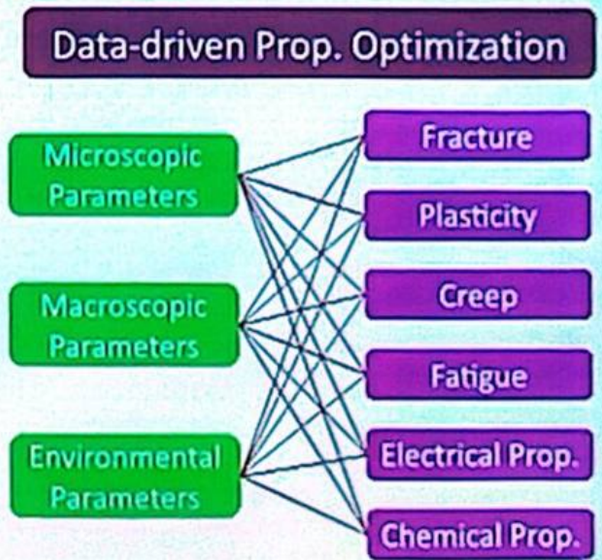


- Data generation →1248 "clean" MD results
  - Absorption, Transmission, Reflection









❑ Key factors

- Character of GB
- Character of ⊥
- Feature size
  (Unit cell, GB thickness)
- GB dislocation
- GB sliding

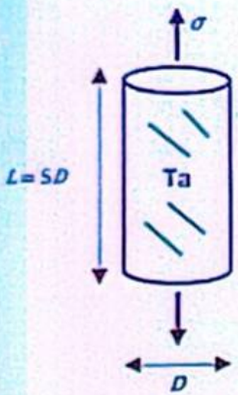
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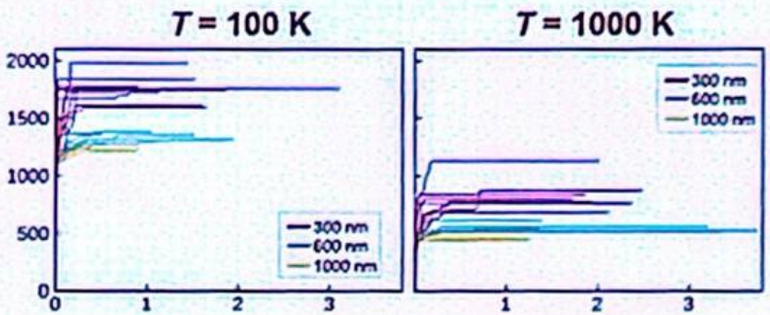
Beyond the Mean Curves

[DD simulations of Ta micropillars]



BCC Ta, [001]

101 DD runs



Size effect
Smaller $D \rightarrow$ higher stress

Thermal softening
Higher $T \rightarrow$ lower stress

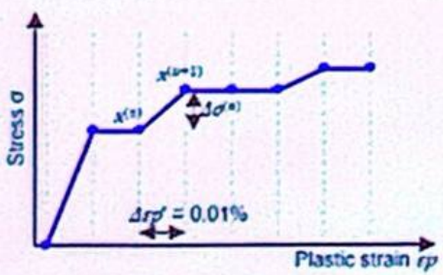
Stochasticity
Smaller $D \rightarrow$ wider scatter

[Hierarchical SVGP emulator : one step of the incremental surrogate]

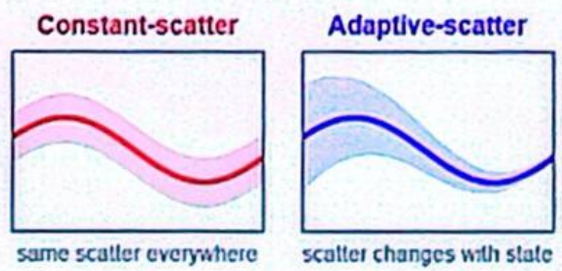
State $x^{(n)}$
$\sigma, \varepsilon_p, \rho, T, D$
→ GP classifier
slip activated?
→ Variance regressors
Var[Ψ(Δσ)], Var[Δρ]
→ GP regressors
Δσ, Δρ
→ Update
$n \rightarrow n+1$

slip → $\Delta\sigma = 0$ | Adaptive-scatter model only | warp $\Psi(\Delta\sigma) = \ln \Delta\sigma + \Delta\sigma/\lambda$

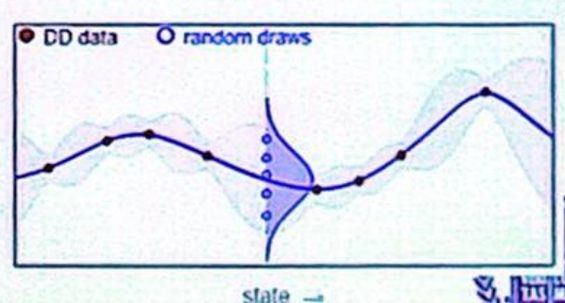
Incremental update



Two models compared



Random draws from the GP



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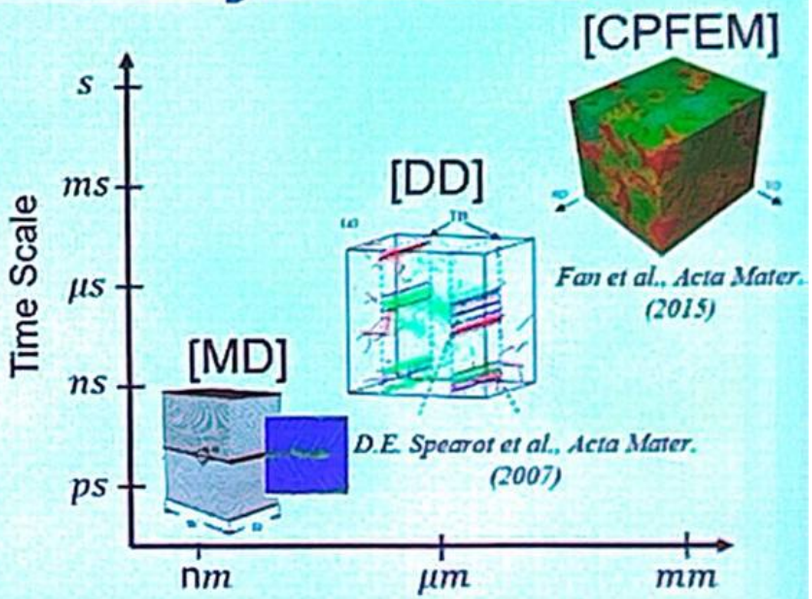
Polycrystal Plasticity

- Characterization of unit interaction

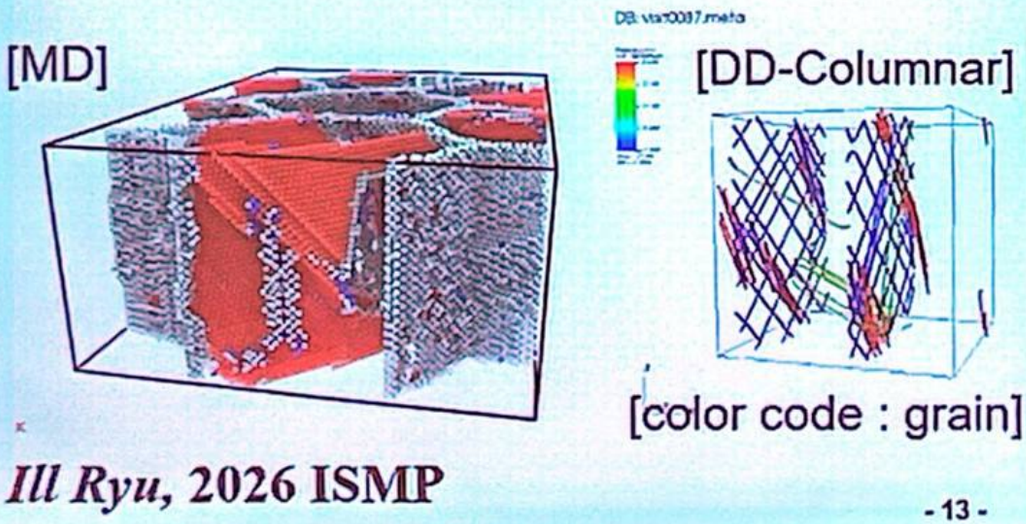
Incoming | Inside GB | Outgoing

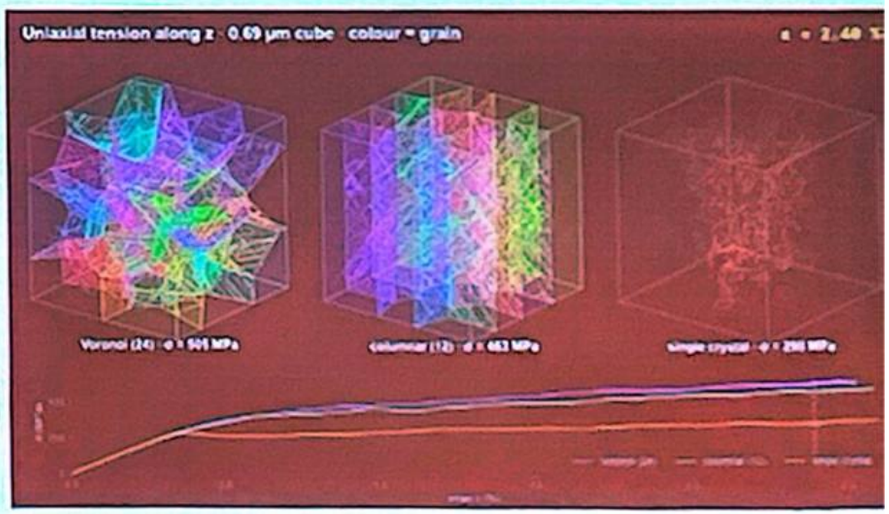
[MD] $[110](\bar{1}11)$ $[110](1\bar{1}1)$ $[10\bar{1}](111)$ $[2\bar{1}1](11\bar{1})$

[DD] $[110](\bar{1}11)$ $[\bar{1}10](1\bar{1}1)$ $[10\bar{1}](111)$ $[101](11\bar{1})$



- DD model with Triple Junctions



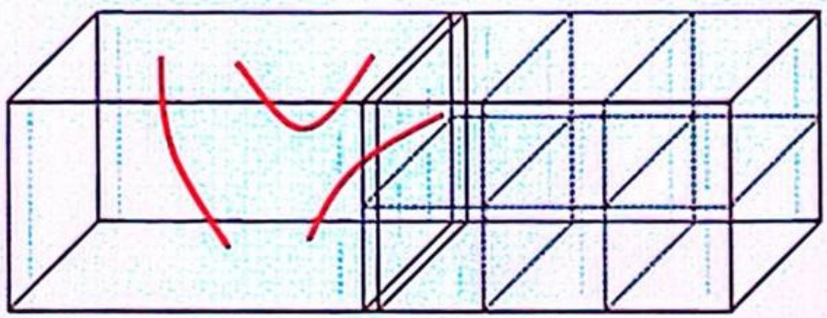


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Defect Dynamics Model

[DDM]

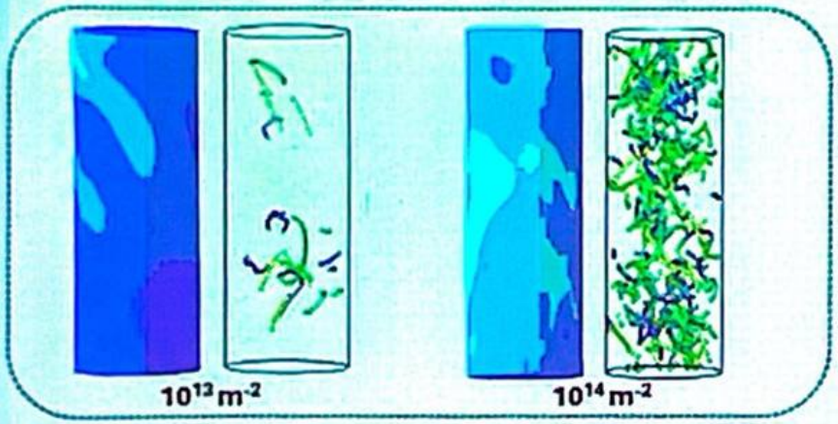


- Accurate plasticity(interaction)
- Arbitrary geometry
- Large deformation
- Multi-physics: Heat, diffusion, damage

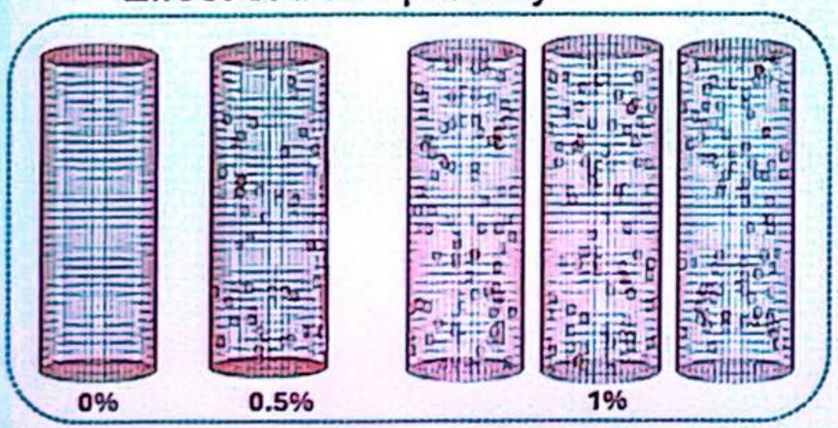
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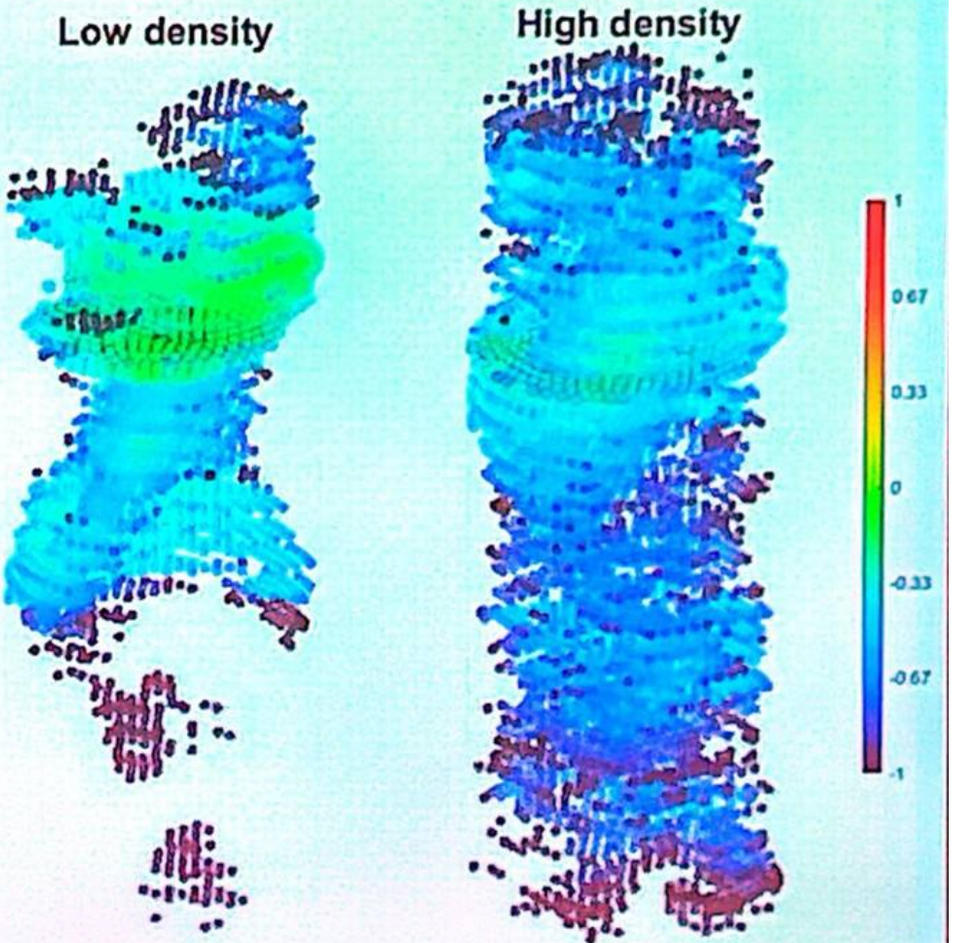
Failure/Damage Prediction using DDM

- Effect of initial dislocation density



- Effect of initial porosity





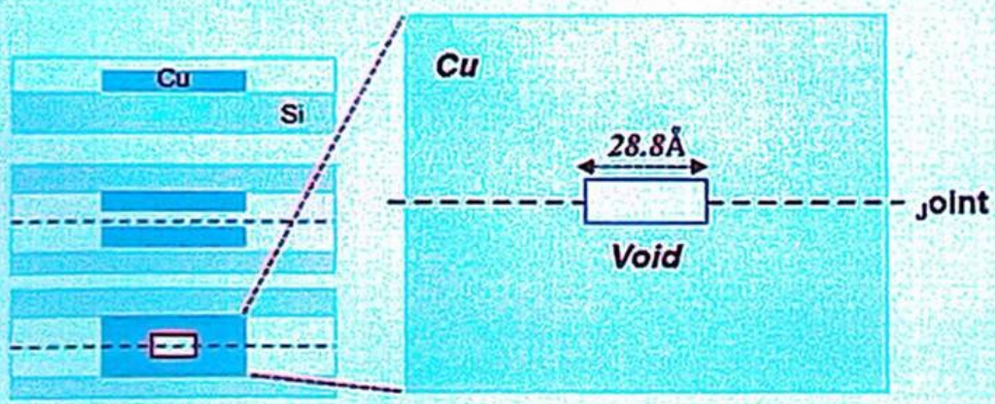
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Orientation Dependence on Cu-Cu bonding

[Model Description]

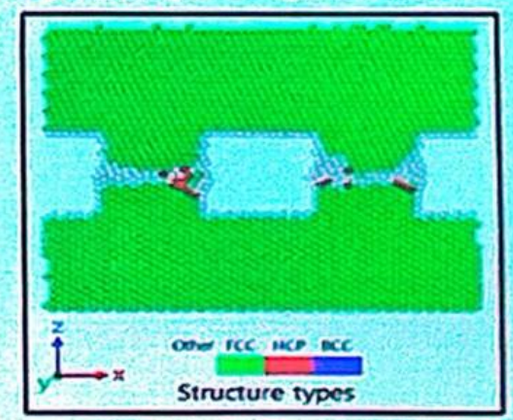
Replaces micro-bump and solder



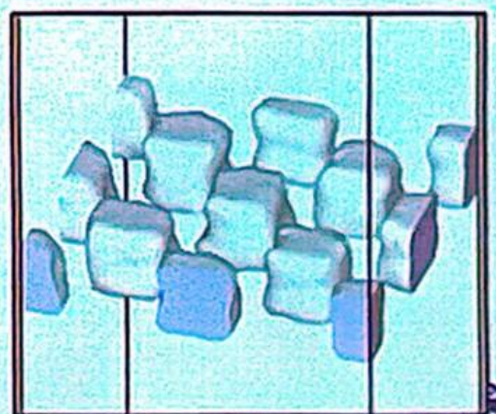
J. De Messemaeker et al., ECTC (2023)
S. Van Huylenbroeck et al., IRPS (2026)

[Bonding Behavior]

{110}/{111} Bonding Interface

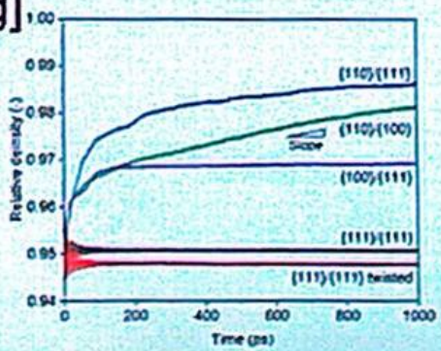


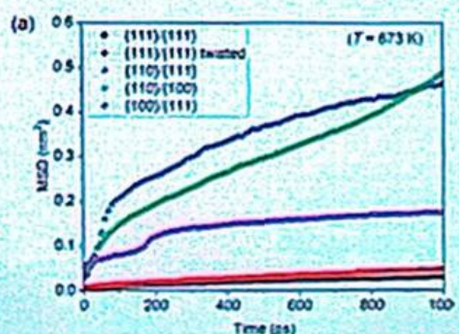
[Bonding Behavior]



[Effect of crystallographic orientation on Cu-Cu bonding]

Relative Density & MSD Curves





H. Tatsumi et al., Sci. Rep. (2023)

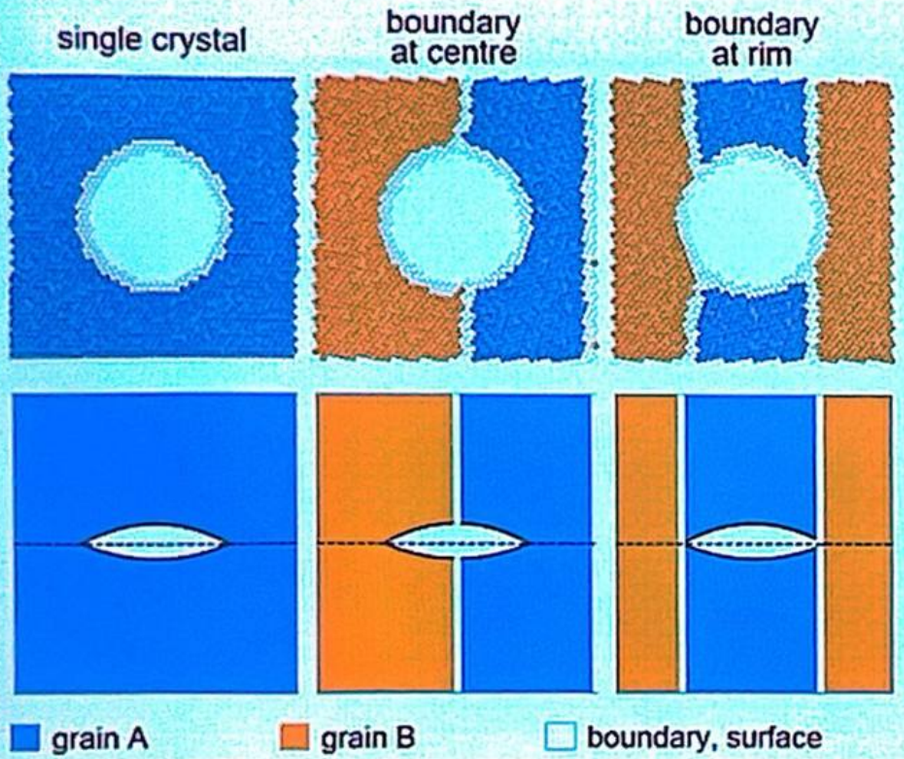
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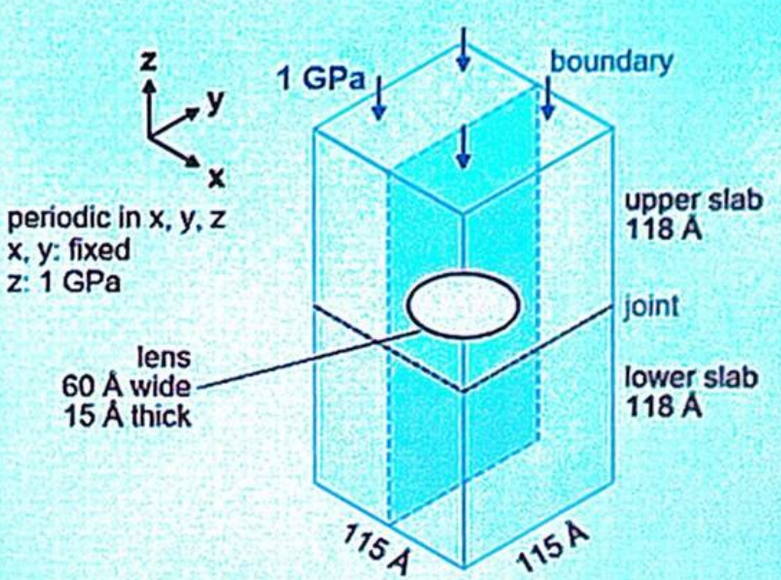
Measuring Void Closing at Grain Boundaries

[Three cells, top and side views]



top: atom layer next to the joint
side: cut through the void centre

[Cell, load and steps]



press	heat	hold
300 K · 50 ps	50 K/ps	673 K · 1–5 ns

Cu EAM: Y. Mishin et al., Phys. Rev. B 63, 224106 (2001)
2.6 × 10⁵ atoms

Hypothesis: a grain boundary can feed atoms to the void, so the void should close faster than in a single crystal.

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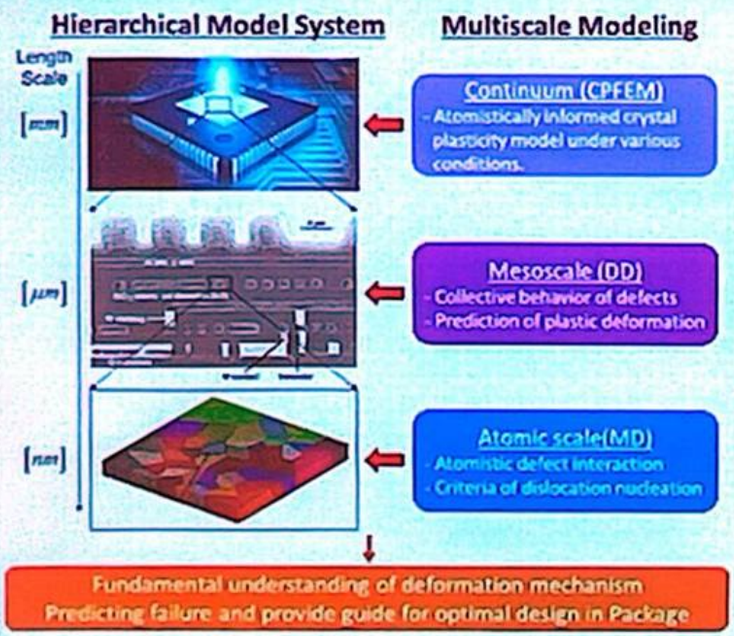
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Integrated Multiscale Defect Engineering

Scientific Impact

Impact1: Microscopic governing mechanism
Impact2 : Understand macroscopic properties

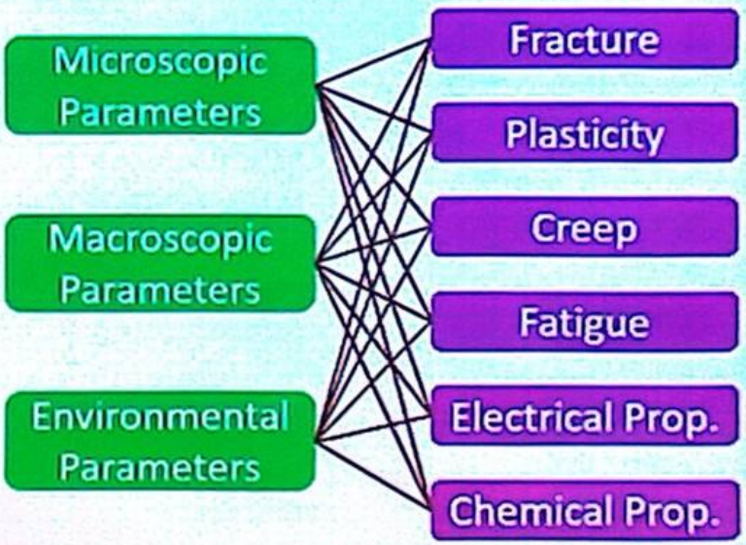
Integrated Defect Engineering



Reliability Analysis

Prediction to Failure through microscopic deformation mechanisms (defects dynamics)

Optimal Performance Design



Ryu, 2026 ISMP